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2023 YEAR 11 EXAMINATION

MATHEMATICS STANDARD 2

MARKING GUIDELINES

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ADVICE FOR TEACHERS

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2023 Year 11 Examination

Mathematics Standard

Marking Guidelines

Section I

Multiple-Choice Answer Key

Question	Answer
1	B
2	D
3	B
4	D
5	A
6	B
7	A
8	C
9	D
10	B
11	B
12	A
13	B
14	C
15	D

Section II

Question 16

Criteria	Marks
• Provides correct solution	2
• Provides correct angle	1

Sample answer

$$\begin{aligned} \text{Area} &= \frac{90}{360} \times \pi \times 7^2 \\ &= 38.4845\dots \\ &= 38 \text{ cm}^2 \end{aligned}$$

Question 17

Criteria	Marks
• Provides correct solution	2
• Provides correct standard voice call/phone charge time	1

Sample answer

$$\begin{aligned} \text{Total charge} &= (\$0.40 \times 8) + (\$0.97 \times 4 \times 8) \\ &= \$34.24 \end{aligned}$$

Question 18

Criteria	Marks
• Provides correct solution	2
• Provides the yearly income	1

Sample answer

$$\text{Salary} = \$2\,875 \times 12 = \$34\,500$$

$$\text{New salary} = \$34\,500 \times 103.5\% = \$35\,707.50$$

Question 19

19 (i)

Criteria	Mark
Provides correct solution	1

Sample answer

Probability of not being caught = 0.6

19 (ii)

Criteria	Marks
Provides correct solution	2
Provides correct probability for one occasion	1

Sample answer

Probability = $(0.6 \times 0.4) + (0.6 \times 0.4) = 0.48$

Question 20

20 (i)

Criteria	Mark
Provides correct solution	1

Sample answer

$P = 24 + 24 + 12 + 12 + 12 + 12 = 96 \text{ cm}$

20 (ii)

Criteria	Marks
Provides correct solution	2
Provides area for one shape of the composite figure	1

Sample answer

Area = $(24 \times 12) + (12 \times 12)$
 $= 432 \text{ cm}^2$

20 (iii)

Criteria	Mark
Provides correct solution	1

Sample answer

$$\text{Volume} = 432 \times 20 = 8640\text{cm}^3$$

Question 21

Criteria	Marks
Provides correct solution	2
Provides the correct ratio of Year 12 students	1

Sample answer

$$\text{Total number of students} = 100 + 120 + 150 + 140 + 110 + 100 = 720$$

$$\text{Students} = \frac{100}{720} \times 50 = 7 \text{ students}$$

Question 22

Criteria	Marks
Provides correct solution	2
Provides correct expansion of brackets	1

Sample answer

$$\begin{aligned} 5(2y - 4) - 3(2y - 5) &= 10y - 20 - 6y + 15 \\ &= 4y - 5 \end{aligned}$$

Question 23

Criteria	Marks
• Provides correct solution	2
• Provides the correct value for the height	1

Sample answer

$$A \approx \frac{4}{2}(1.5 + 1.8) + \frac{4}{2}(1.8 + 1.5)$$
$$\approx 13.2 \text{ m}^2$$

Question 24

Criteria	Marks
• Provides correct solution	2
• Provides correct total degrees of difference	1

Sample answer

$$\text{Total degrees} = 24 + 47 = 71 \text{ degrees}$$

$$\text{Time} = \frac{71}{15} = 4 \text{ hours } 44 \text{ minutes}$$

Question 25

Criteria	Marks
• Provides correct solution	2
• Insert marking criteria	1

Sample answer

$$\text{Stamp duty} = (\$54\,200 - \$45\,000) \times 5\% + \$1\,350$$
$$= \$1\,810$$

Question 26

Criteria	Marks
• Provides correct solution	2
• Provides correct one year increase	1

Sample answer

$$\text{After 1st year} = \$55\,000 \times 1.02 = \$56\,100$$

$$\text{After 2nd year} = \$56\,100 \times 1.03 = \$57\,783$$

Question 27

Criteria	Marks
• Provides correct solution	2
• Provides correct total frequency	1

Sample answer

$$\text{Mean} = \frac{(0 \times 5) + (1 \times 8) + (2 \times 3) + (3 \times 2)}{18} = 1.11 \dots$$

Hence, 1 brother.

Question 28

28 (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct area for triangle	1

Sample answer

$$\text{Area} = \left(\frac{1}{2} \times 4 \times 3.5 \right) \times 6 = 42 \text{ cm}^2$$

28 (ii)

Criteria	Marks
• Provides correct solution	2
• Provides correct volume in volume units	1

Sample answer

$$\text{Volume} = 42 \times 25 = 1\,050 \text{ cm}^3$$

$$\text{Capacity} = 1050 \div 1000 = 1 \text{ L}$$

Question 29

Criteria	Marks
• Provides correct solution	2
• Provides correct time difference	1

Sample answer

$$\text{Time difference} = 16 \text{ hours}$$

$$\text{Time in Chicago} = 8 \text{ pm Thursday} + 3 \text{ hours} = 11 \text{ pm Thursday}$$

Question 30

30 (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct tax bracket	1

Sample answer

$$\text{Tax payable} = ((\$98\,750 - \$45\,000) \times 0.325) + \$5\,092 = \$22\,560.75$$

30 (ii)

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\text{Medicare levy} = \$98\,750 \times 2\% = \$1\,975$$

30 (iii)

Criteria	Marks
• Provides correct solution	2
• Provides correct total tax owed	1

Sample answer

$$\text{Total tax owed} = \$22\,560.75 + \$1\,975 = \$24\,535.75$$

$$\text{Tax refund} = \$28\,080 - \$24\,535.75 = \$3\,544.25$$

Hence, Anton will receive a refund of \$3 544.25

Question 31

Criteria	Marks
• Provides correct solution	2
• Provides correct kilojoules burnt for incorrect weight	1

Sample answer

$$\text{Total kilojoules} = \left(\left(\frac{40}{60} \times 23 \right) + \left(\frac{10}{60} \times 22.1 \right) + \left(\frac{70}{60} \times 34.1 \right) \right) \times 55 = 3234 \text{ kJ}$$

Question 32

Criteria	Marks
• Provides correct solution	2
• Provides correct moving of one part of equation	1

Sample answer

$$\frac{x}{2} + 3 = 12$$

$$\frac{x}{2} = 9$$

$$x = 18$$

Question 33

Criteria	Marks
• Provides correct solution	3
• Provides correct value of k	2
• Provides correct equation	1

Sample answer

$$t = \frac{k}{s^2}$$

$$\frac{20}{60} = \frac{k}{4^2}$$

$$k = \frac{16}{3}$$

$$t = \frac{16}{3s^2}$$

$$t = \frac{16}{3 \times 8^2}$$

$$t = \frac{1}{12} = 5 \text{ minutes}$$

Question 34

34 (i)

Criteria	Mark
Provides correct solution	1

Sample answer

$$C = \frac{5}{9}(75 - 32) = 24^{\circ}C$$

34 (ii)

Criteria	Marks
Provides correct solution	2
Provides correct moving of one part of equation	1

Sample answer

$$C = \frac{5}{9}(F - 32)$$

$$9C = 5(F - 32)$$

$$\frac{9C}{5} = F - 32$$

$$F = 32 + \frac{9C}{5}$$

34 (iii)

Criteria	Mark
Provides correct solution	1

Sample answer

$$F = 32 + \frac{9C}{5}$$

Let C be 40.

$$F = 32 + \frac{9 \times 40}{5} = 104^{\circ}F$$

Question 35

35 (i)

Criteria	Mark
Provides correct solution	1

Sample answer

$$\text{Total weight} = (40 \times 40) + (10 \times 30) = 1900 \text{ kg}$$

35 (ii)

Criteria	Marks
Provides correct solution	2
Provides correct y-intercept	1

Sample answer

$$\begin{aligned}
 N &= -\frac{48}{60}n + 48 \\
 &= -\frac{4}{5}n + 48
 \end{aligned}$$

35 (iii)

Criteria	Mark
Provides correct solution	1

Sample answer

$$N = -\frac{4}{5} \times 20 + 48 = 32$$

35 (iv)

Criteria	Marks
• Provides correct solution	2
• Provides correct description of y-intercept	1

Sample answer

The y – intercept represents how many 40 kg drums there are initially.

The gradient is the change in 40 kg drums per 30 kg drum.

Question 36

Criteria	Marks
• Provides correct solution	3
• Provides correct charge for non-peak rate	2
• Provides correct charge for peak rate	1

Sample answer

$$\text{Saving} = (\$0.26334 - \$0.12199) \times 0.5 \times 365 \times 1.2 = \$30.96$$

Question 37

Criteria	Marks
• Provides correct solution	3
• Provides correct interest charge	2
• Provides correct monthly interest rate	1

Sample answer

$$\text{Interest} = 12\,000 \times \frac{14.5}{12} \% \times 3 = \$435$$

$$\text{Owed amount} = \$12\,000 + \$435 = \$12\,435$$

Question 38

Criteria	Marks
• Provides correct solution	2
• Provides correct substitution	1

Sample answer

$$0.05 = \frac{(10 \times 6 - 7.5H)}{5.5 \times 55}$$

$$15.125 = 60 - 7.5H$$

$$-44.875 = -7.5H$$

$$H = 6 \text{ hours (to the nearest hour)}$$

Question 39

Criteria	Marks
• Provides correct solution	2
• Provides correct probability for first red jellybean	1

Sample answer

$$\text{Probability} = \frac{4}{15} \times \frac{3}{14} = \frac{12}{210} = \frac{2}{35}$$

2023 Year 11 Mathematics Standard 2 Examination

Mapping Grid

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Substitution	MS-A1
2	1	Complementary event	MS-S2
3	1	Percentage error	MS-M1
4	1	Scientific notation	MS-M1
5	1	Time difference	MS-M2
6	1	Absolute error	MS-M1
7	1	Pareto chart	MS-S1
8	1	Five-number summary	MS-S1
9	1	GST	MS-F1
10	1	Straight-line depreciation	MS-F1
11	1	Equation of a line	MS-A2
12	1	Income	MS-F1
13	1	Area	MS-M1
14	1	Fuel consumption	MS-F1
15	1	Numerical data	MS-S1

Section II

Question	Marks	Content	Syllabus Outcomes
16	2	Area of sector	MS-M1
17	2	Budgeting	MS-F1
18	2	Salary	MS-F1
19 (i)	1	Complementary event	MS-S2
19 (ii)	2	Multi-stage event	MS-S2
20 (i)	1	Perimeter	MS-M1
20 (ii)	2	Area	MS-M1
20 (iii)	1	Volume	MS-M1
21	2	Stratified sample	MS-S1
22	2	Algebraic expressions	MS-A1
23	2	Trapezoidal rule	MS-M1
24	2	Latitude and longitude	MS-M2
25	2	Stamp duty	MS-F1
26	2	Percentage increase	MS-F1
27	2	Mean	MS-S1
28 (i)	2	Area	MS-M1
28 (ii)	2	Capacity	MS-M1
29	2	Local time	MS-M2
30 (i)	2	Tax payable	MS-F1
30 (ii)	1	Medicare levy	MS-F1

Question	Marks	Content	Syllabus Outcomes
30 (iii)	2	Tax refund	MS-F1
31	2	Energy	MS-M1
32	2	Solving an equation	MS-A1
33	3	Inversely proportional	MS-A2
34 (i)	1	Substitution	MS-A1
34 (ii)	2	Change of subject	MS-A1
34 (iii)	1	Substitution	MS-A1
35 (i)	1	Linear relationships	MS-A2
35 (ii)	2	Equation of a line	MS-A2
35 (iii)	1	Substitution	MS-A1
35 (iv)	2	Gradient and intercept	MS-A1
36	3	Energy	MS-M1
37	3	Simple interest	MS-F1
38	2	Blood and alcohol	MS-A1
39	2	Multi-stage probability	MS-S2